visibility timeout for other workers. This is called a *retry*. But if for some reason every retry fails (maybe you have a bug in your code), the message will reside in the queue forever and may waste a lot of resources because of all the retries. To avoid this, you can configure a dead-letter queue. If a message is retried more than a specific number of times, it's removed from the original queue and forwarded to the DLQ. The difference is that no worker listens for messages on the DLQ. You should create a CloudWatch alarm that triggers if the DLQ contains more than zero messages, because you need to investigate this problem manually by looking at the message in the DLQ. Once the bug is fixed, you can move the messages from the dead letter queue back to the original queue to process them again.

Now that the basic resources have been designed, let's move on to the more specific resources.

IAM ROLES FOR SERVER AND WORKER EC2 INSTANCES

Remember that it's important to grant only the privileges that are necessary. All server instances must be able to do the following:

- `sqs:SendMessage` to the SQS queue created in the template to trigger image processing
- `s3:PutObject` to the S3 bucket created in the template to upload a file to S3 (You can further limit writes to the `upload/` key prefix.)
- `dynamodb:GetItem`, `dynamodb:PutItem`, and `dynamodb:UpdateItem` to the DynamoDB table created in the template

All worker instances must be able to do the following: